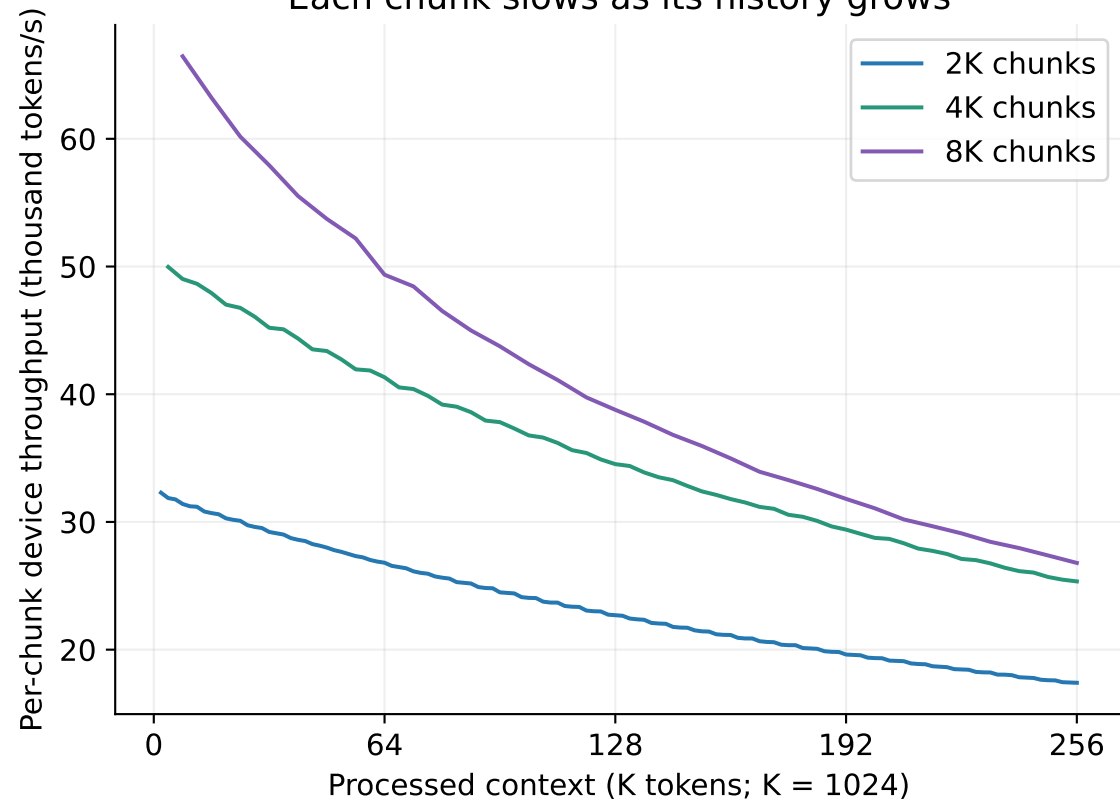
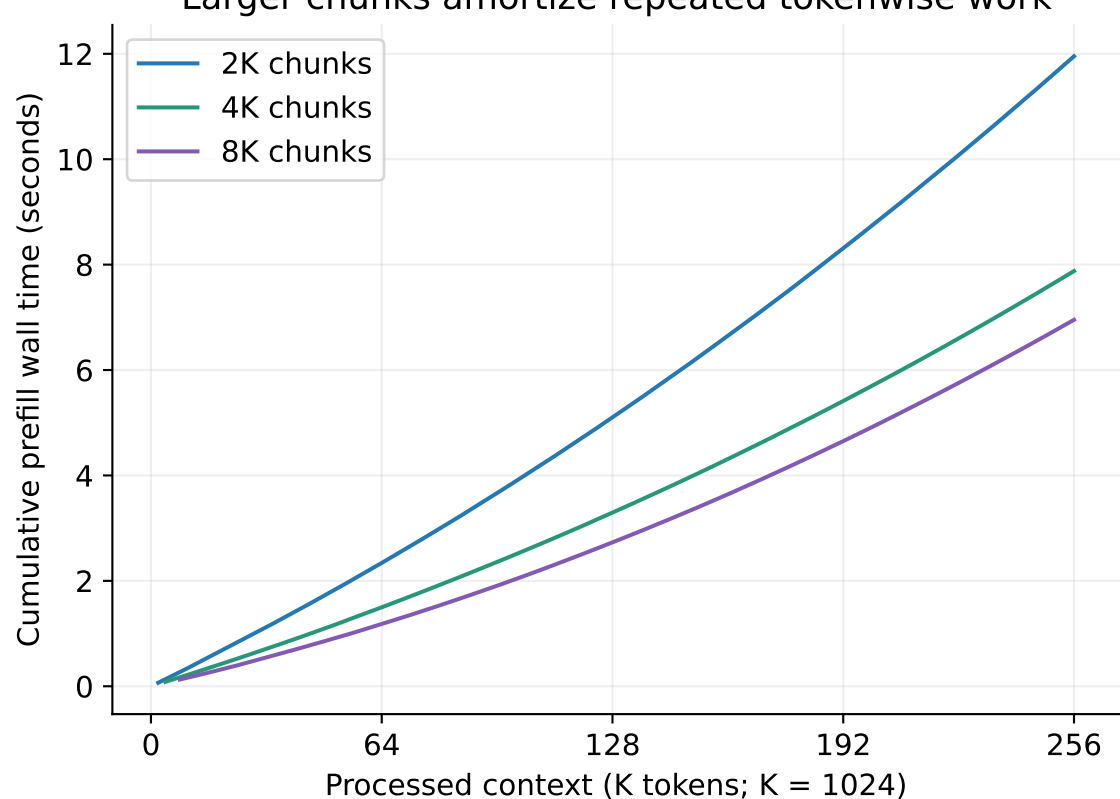


Each chunk slows as its history grows



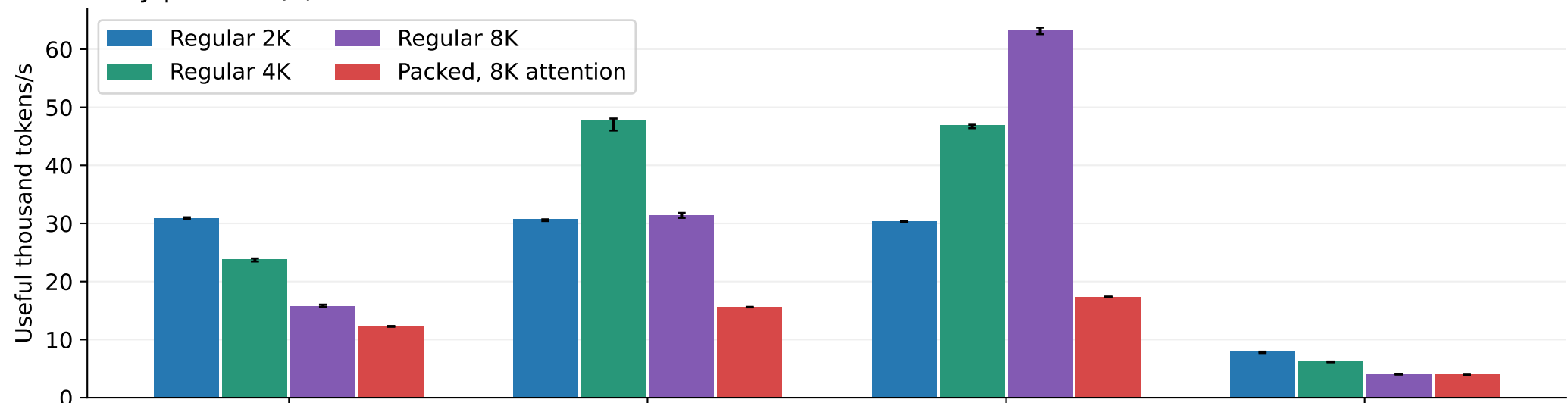
Larger chunks amortize repeated tokenwise work



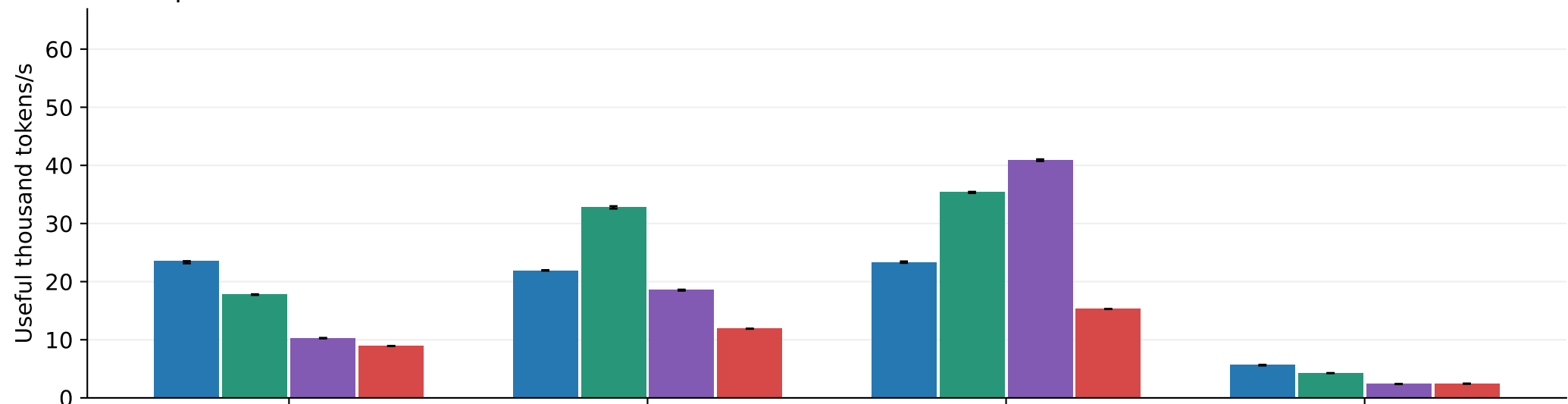
Why: Establish the best ordinary prefill path and retain its complete context curve.

Finding: These are the requested unmodified 256K canonical tests. The left panel uses printed device timing; the right includes staging. At the last chunk, 4K retains 94.6% of 8K's device throughput, with 161.6 ms versus 305.8 ms per chunk. Smaller chunks provide shorter scheduling steps, but 8K remains fastest over the entire prompt. Every printed chunk is preserved in canonical_chunks.csv.

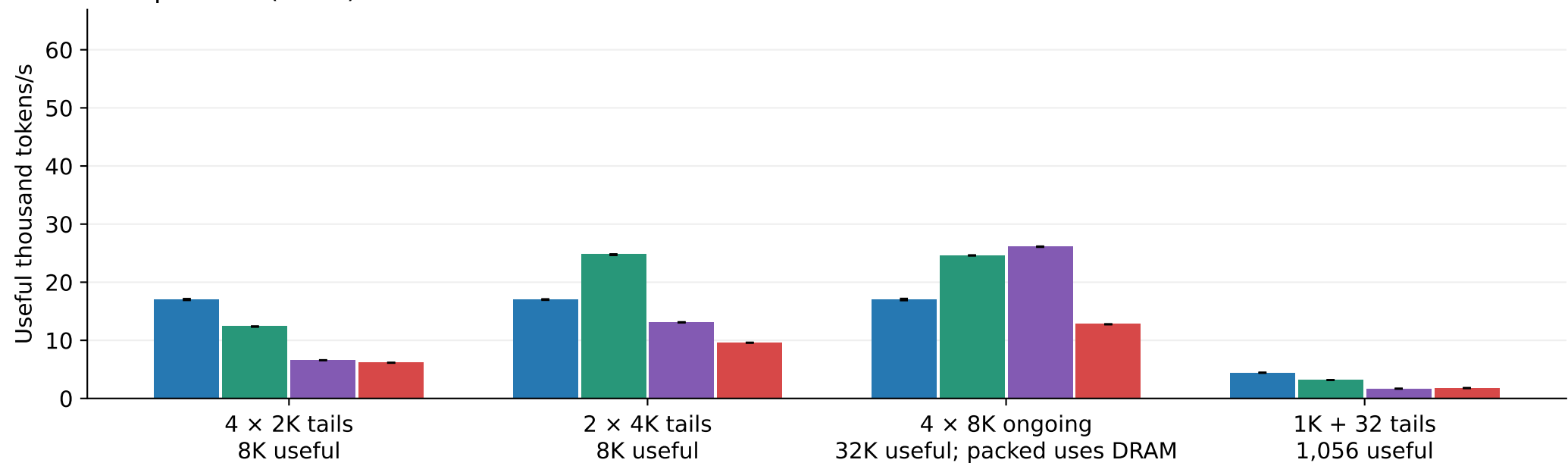
Early prefixes (0)



Mixed prefixes



Late prefixes (248K)

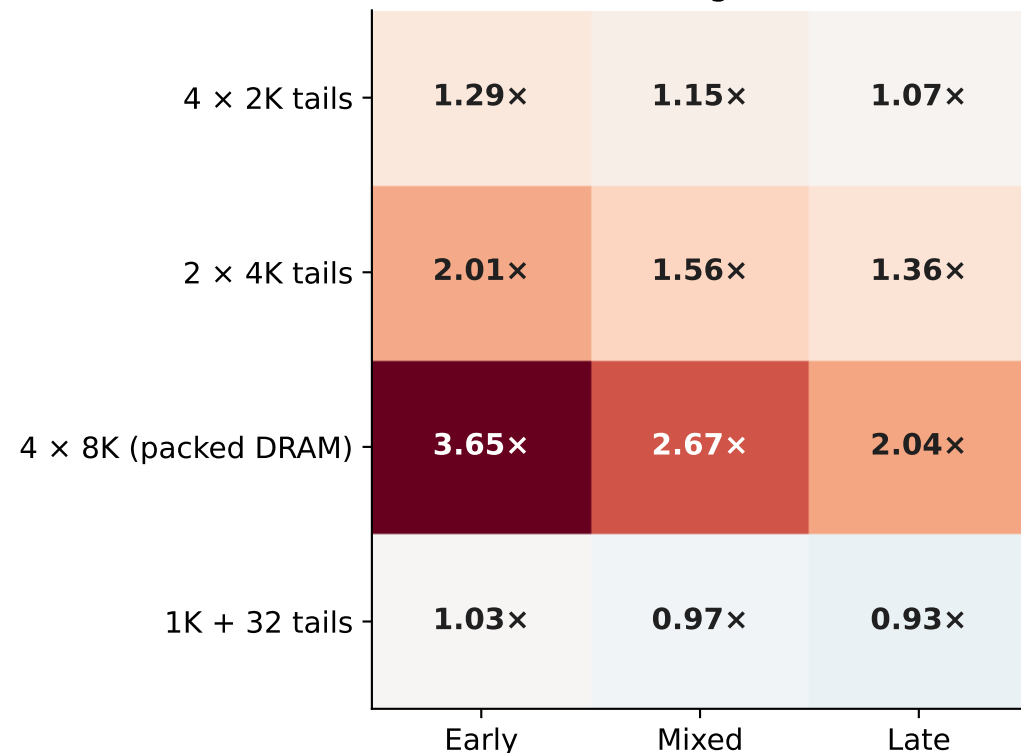


Why: Compare real multi-slot batches against all three native chunk sizes, including final tails and full ongoing chunks.

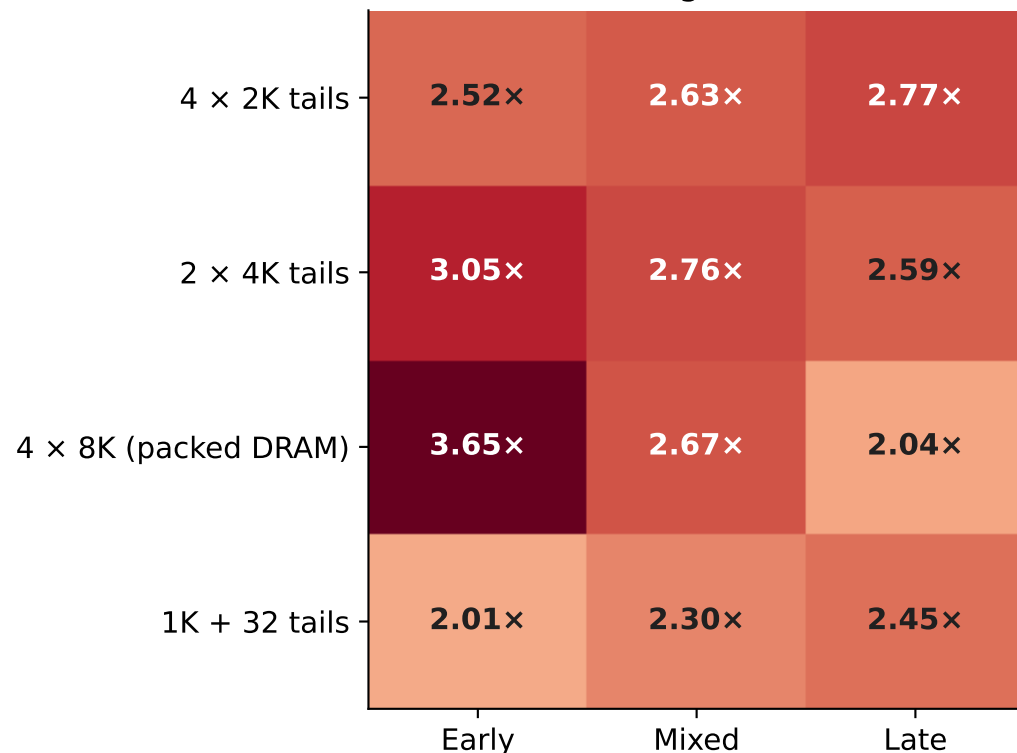
Finding: Five-replay medians; whiskers show the observed min-max range. Early = start 0; late = start 248K; mixed = [0, 8K, 128K, 248K] for four requests and [0, 248K] for two. Hatched full-batch packed bars use the explicit DRAM-activation override: the default L1 path failed at this size. Other bars use default activation placement. Packing retains 8K attention per request.

Packed batch time ÷ native batch time (>1× means slower)

Versus regular 8K



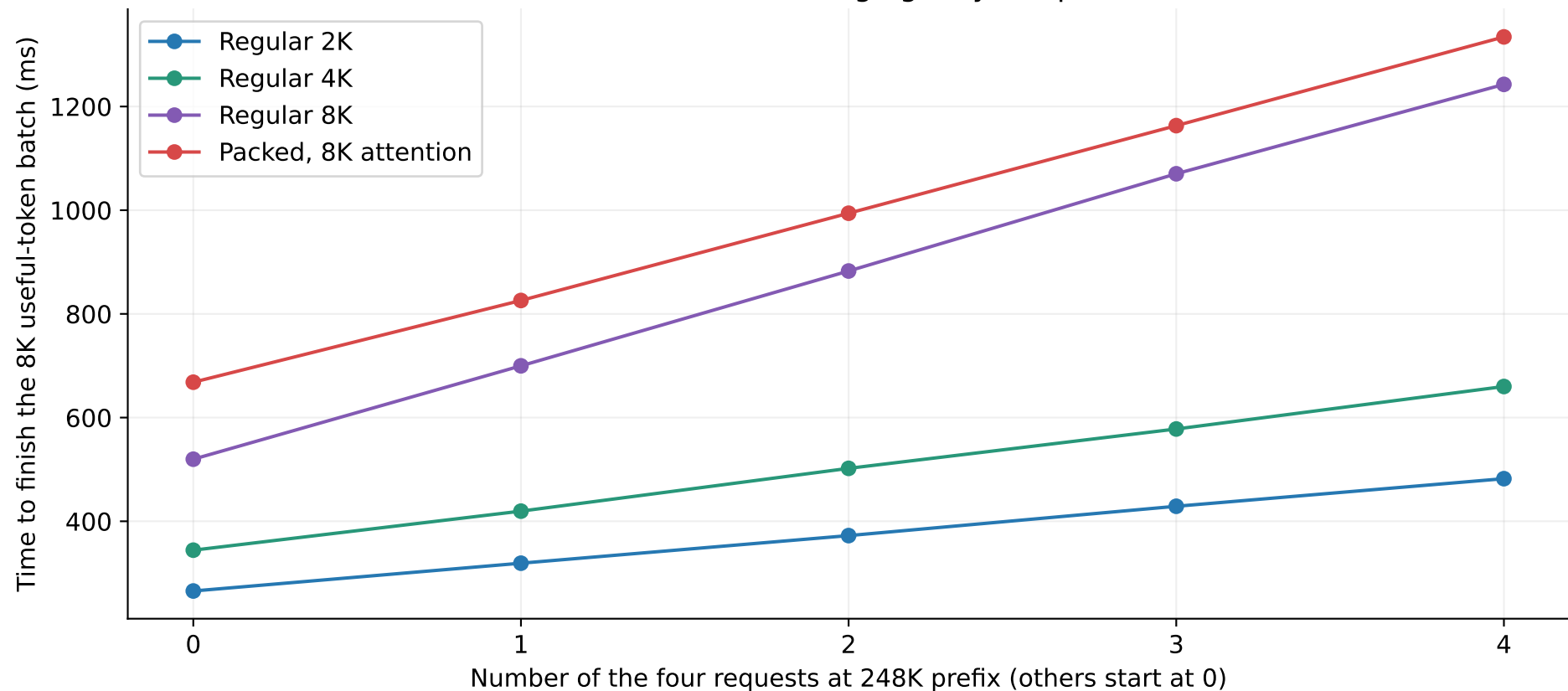
Versus best regular chunk size



Why: Separate comparison with the current 8K default from comparison with an appropriately chosen native chunk size.

Finding: Cells are packed batch time ÷ native batch time. Below 1× favors packing (blue); above 1× means packing is slower (red). Four full packed chunks require the measured DRAM fallback. The best regular chunk is selected separately for each workload; changing chunk size within a live request would require compatible cache geometry, and is not a free scheduler decision.

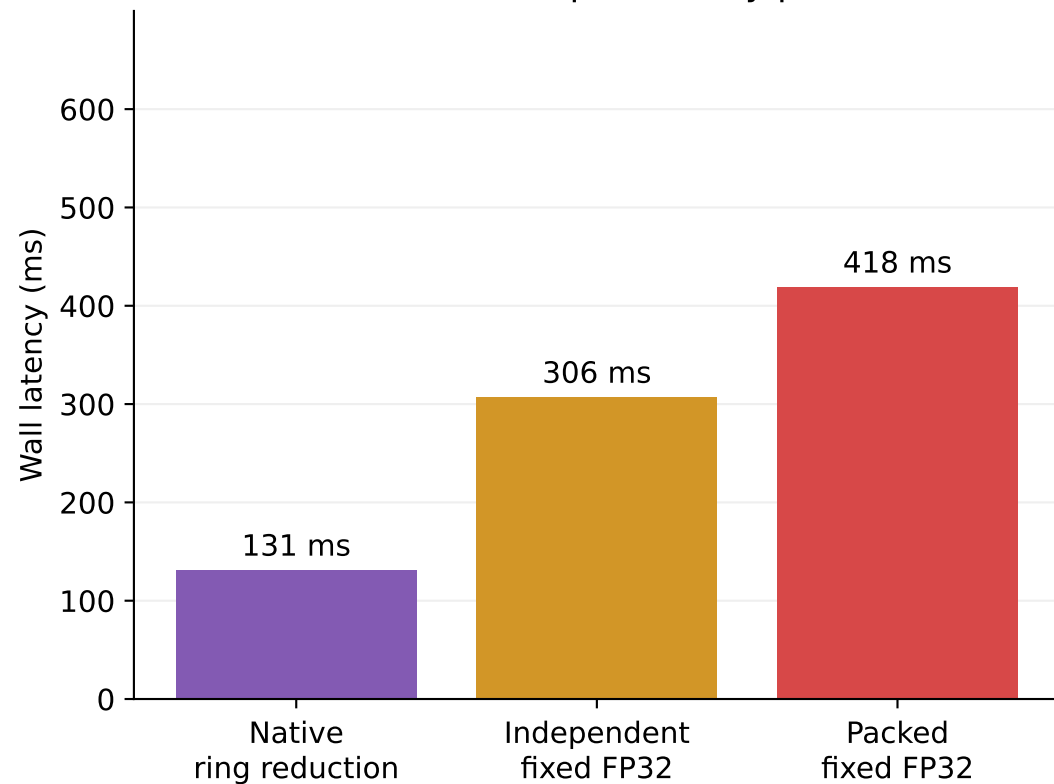
Same four 2K tails, changing only the prefix mix



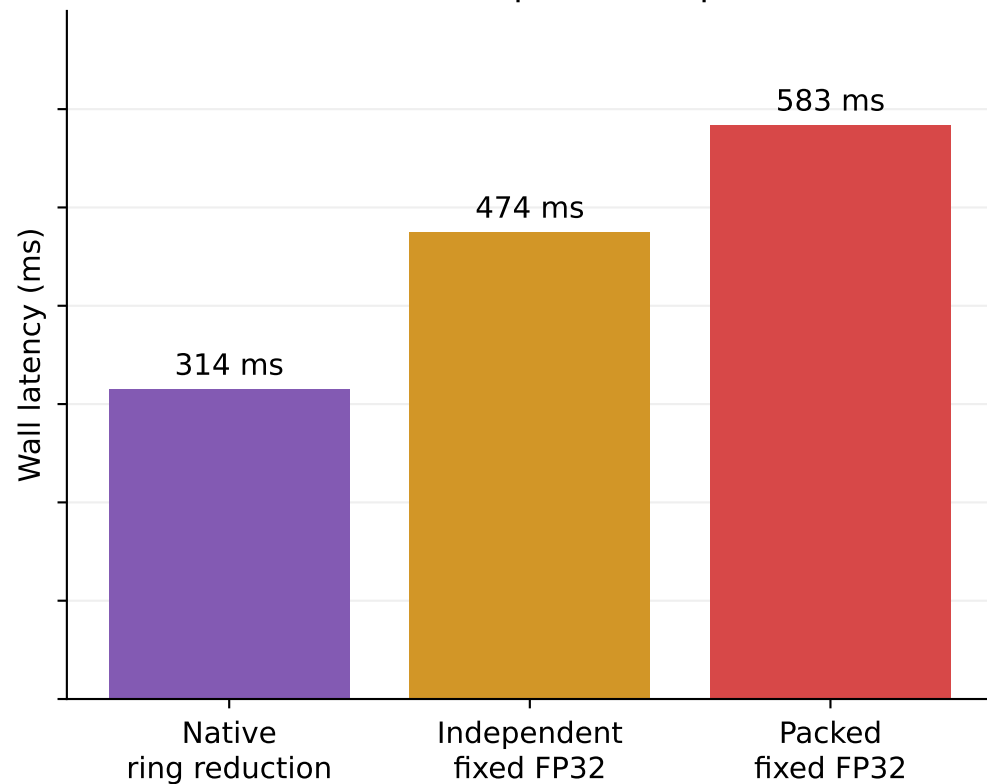
Why: Hold useful tokens, number of requests, and packing shape fixed; vary only how many prefixes are long.

Finding: Moving one request from start 0 to 248K adds about 166.5 ms in the packed path versus 54.2 ms with native 2K chunks (endpoint-average increments). Prefix changes reuse the resident trace, so this increase is execution work, not recapture. Mixed batches pay approximately the sum of their separate history costs, not the longest history for every request. Per-request 8K attention still executes on the packed path, including padded queries; shared projections do not eliminate this history-dependent work.

One full 8K request, early prefix



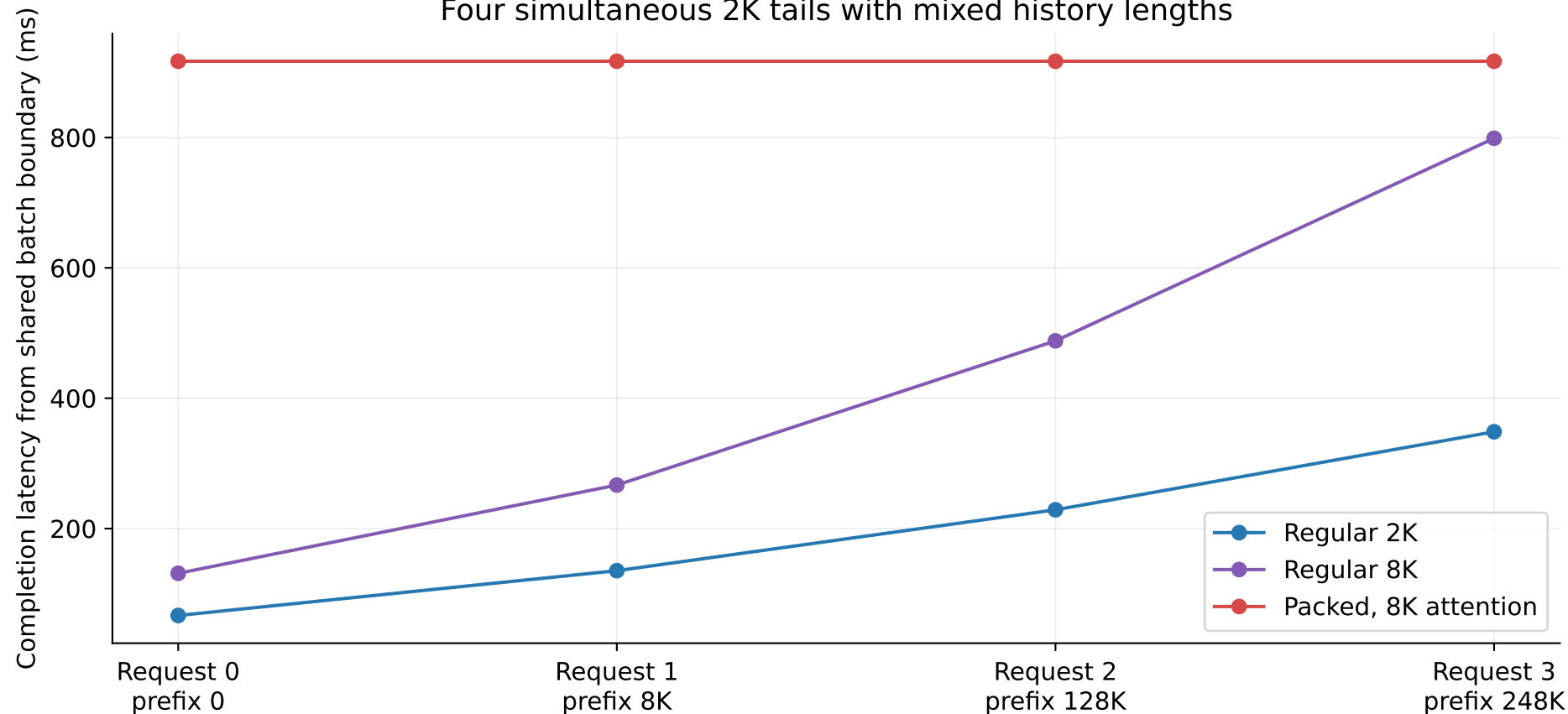
One full 8K request, late prefix



Why: Use a single full request so packing saves no rows, then isolate the stable-reduction choice from the remaining packed machinery.

Finding: The first-to-second difference measures replacing native TP ring reductions with rank-ordered FP32 additions: 176 ms extra at the early prefix. The second-to-third difference includes CP redistribution, splitting/concatenating, and the packed cache/metadata path. For four early 2K tails, packing takes 668 ms versus 1229 ms with independent fixed-FP32 execution, but the default native path takes only 520 ms. The saving against the FP32 control is real; it does not establish a win over the default. This is an end-to-end ablation, not a kernel-profiler attribution. Numerical equivalence to GPU is not established by this performance study.

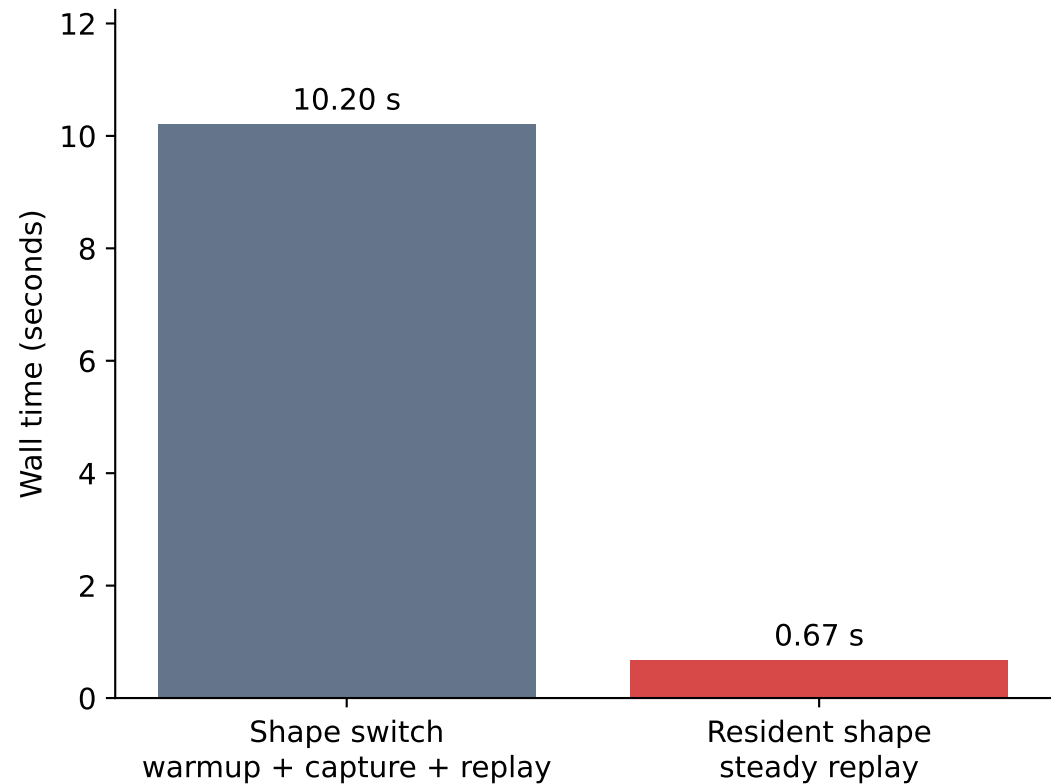
Four simultaneous 2K tails with mixed history lengths



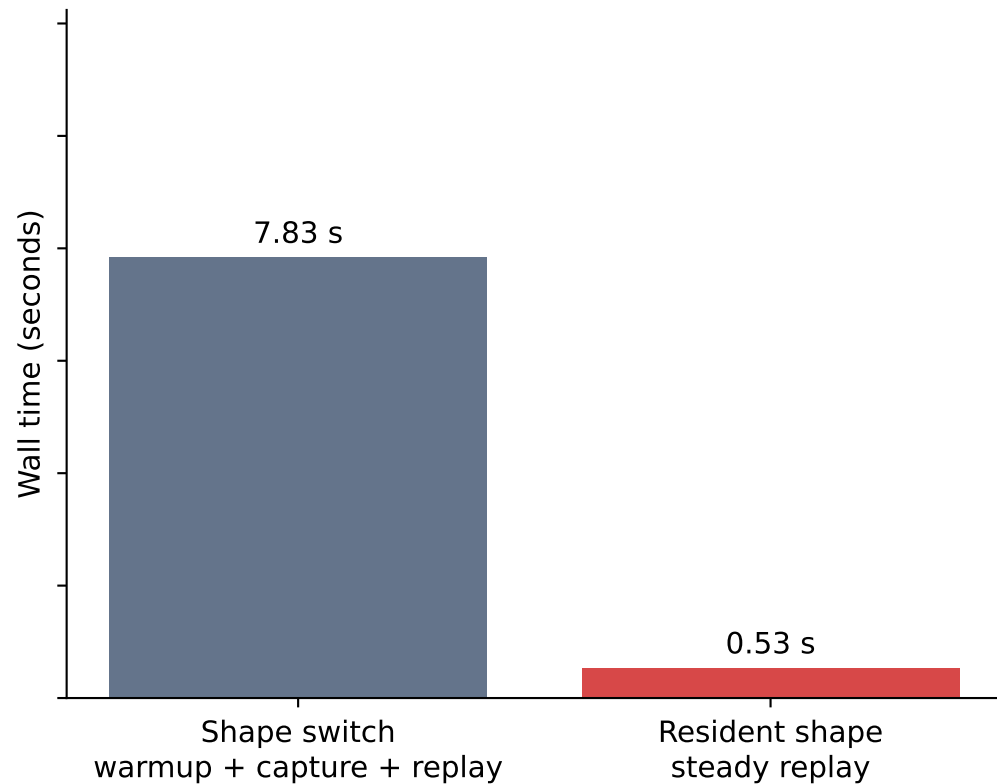
Why: Avoid mistaking aggregate throughput or batch time divided by request count for actual request completion latency.

Finding: All requests are ready at time zero. Native paths visit active slots round-robin by their chunk size; packed outputs become available only after the entire graph completes. Earlier requests therefore inherit the long-history work of other requests in their packed batch. Arrival queueing and downstream transfer would add further latency.

Four 2K tails



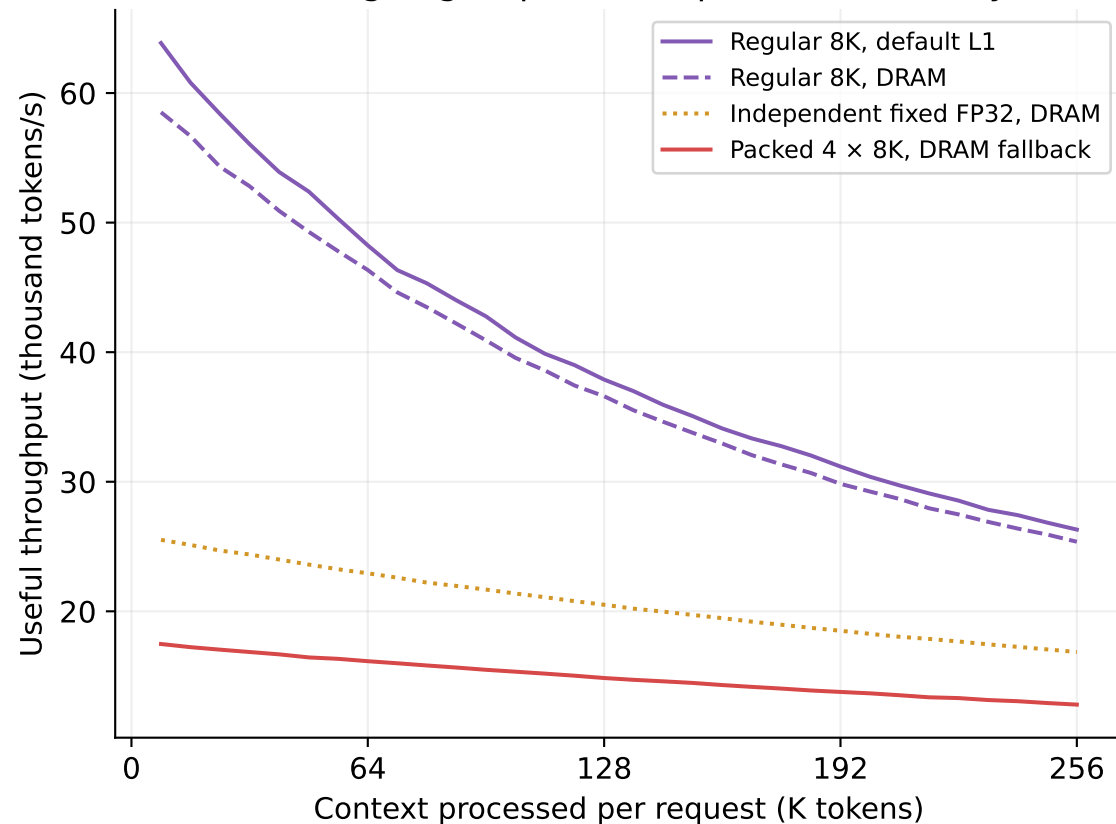
Two 4K tails



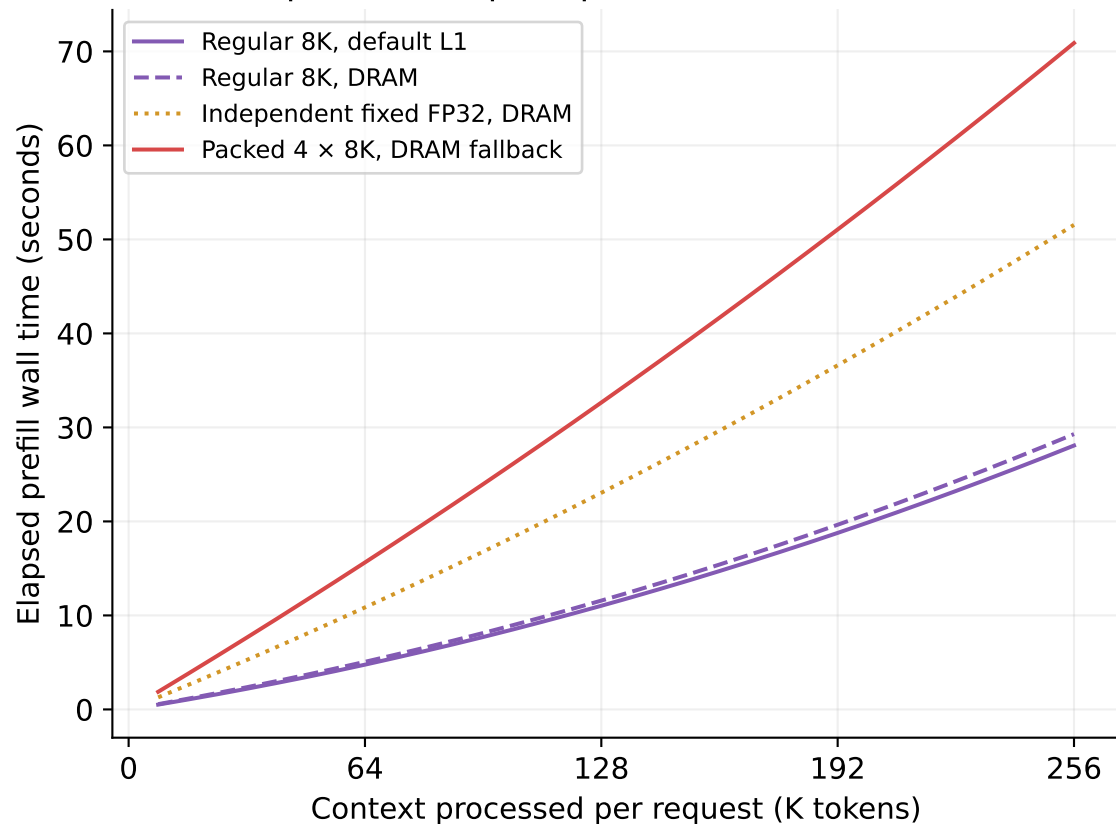
Why: Revisit shapes whose kernels have already run, after their traces have been released.

Finding: These are warm-process shape switches, not weight loading. Only one graph is resident, so returning to an old shape still triggers warmup/capture. Fixed-shape replay throughput is an optimistic bound for a heterogeneous queue; frequent shape changes can dominate service time.

Four ongoing requests keep the device busy



Four complete 256K prompts = 1,048,576 useful tokens



Why: Measure a real continuous prefill stream instead of extrapolating fixed-prefix tail measurements.

Finding: Each path runs all four prompts from 0 to 256K with real contiguous KV writes. Native 8K took 28.07 s (37,353 useful tokens/s); packed with DRAM activations took 70.89 s (14,792 tokens/s). Native 8K with DRAM took 29.29 s; independent fixed-FP32 execution with DRAM took 51.55 s. These controls separate storage placement and reductions from the remaining packing costs. This is one complete stream per mode, excluding initial warmup/capture. The packed default L1 configuration failed before this stream could run.